# Lab 3: Use Case Diagram

System: Library Management System.

Main tool: diagrams.net (draw.io) in a web browser.

No programming/coding is required.

# Aim and expected output

Aim: identify actors and use cases and draw a complete use case diagram.

Use your Lab 1 Library Management System SRS.

At the end, show/save one clear Use Case Diagram.

You should also be able to explain your diagram in the viva.

# Before you start

Open your Lab 1 SRS.

Open a browser and go to diagrams.net (draw.io).

Keep a notebook/Word file for your actor and use-case list.

Do not copy a ready-made diagram from the internet.

# Step 1: Find the actors

Start with Member and Librarian.

Check your SRS for any other external role actually required.

Write the actor names before drawing.

Do not make 'Book' an actor; a book does not interact with the software as a user.

# Step 2: Find the use cases

Member: Search Book, View Availability, Reserve Book.

Librarian: Issue Book, Return Book, Maintain Book Records.

Use your own SRS to add/remove functions if required.

Use action words for use-case names.

# Step 3: Open diagrams.net

Open diagrams.net in the browser.

Choose where you want to save the file, or use the available device option.

Choose 'Create New Diagram' and select a blank diagram.

Give the file a simple name such as LMS_Use_Case_Diagram.

# Step 4: Add UML shapes

Look for UML/UML Use Case shapes in the shape library.

Use the actor/stick-person shape for Member and Librarian.

Use oval shapes for use cases.

Use a large rectangle for the system boundary.

Write 'Library Management System' on the rectangle.

# Step 5: Place everything correctly

Put Member and Librarian OUTSIDE the rectangle.

Put Search Book, Reserve Book, Issue Book etc. INSIDE the rectangle.

Use simple lines to connect actors with the functions they use.

Keep the diagram neat and readable.

# Step 6: Include/Extend only if needed

Do not make the diagram unnecessarily complex

You may add <<include>> or <<extend>> only when your requirement supports it.

Example include: Issue Book -> Check Member Details, if checking is always required.

Example extend: Pay Fine -> Return Book, if payment occurs only when a fine exists.

A correct simple diagram is better than an incorrect complicated diagram.

# Step 7: Check and save

Are all actors outside the system box?

Are all use cases inside?

Does every connection make sense?

Does the diagram agree with your SRS?

# Result to write

Example: 'In this experiment, we prepared a Use Case Diagram for the Library Management System.'

'We identified the main actors and use cases and represented their interactions in the diagram.'